

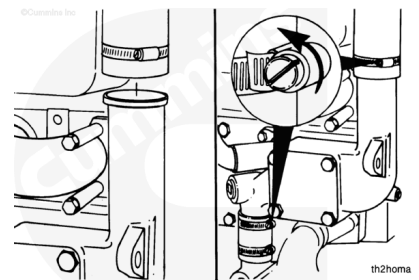
Trainings ▾

Remove

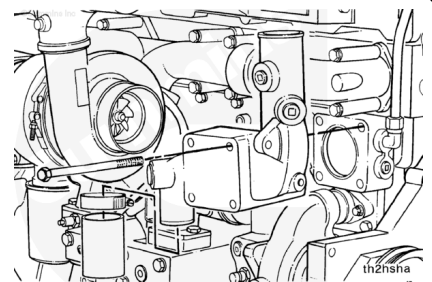
With EGR

Loosen the coolant bypass hose clamps on both the upper and lower hoses.

Remove the upper coolant hose from the thermostat housing.



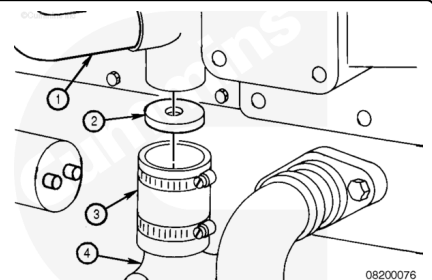
Remove the four thermostat housing mounting capscrews and the thermostat housing.



The coolant flow that provides cooling to the torque converter (if equipped) is achieved in different manners.

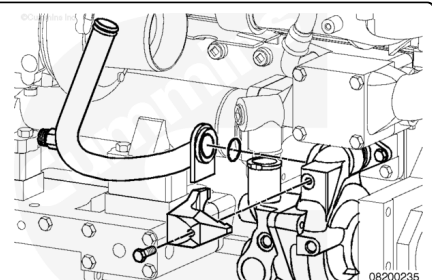
ISM and QSM Series engines use a torque converter cooler disc inside the coolant bypass hose to direct engine coolant to the inlet side of the torque converter cooler.

1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.



Loosen the EGR cooler coolant supply lower tube retaining clamp capscrew. Remove the EGR cooler coolant supply lower tube retaining clamp and tube.

Remove and discard the o-ring.

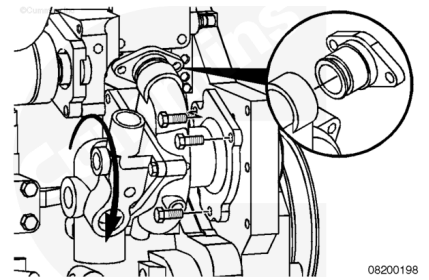


Remove the three water pump mounting capscrews.

Remove the two water pump water transfer connection capscrews and push the water pump transfer connection into the water pump as far as possible to facilitate water pump removal.

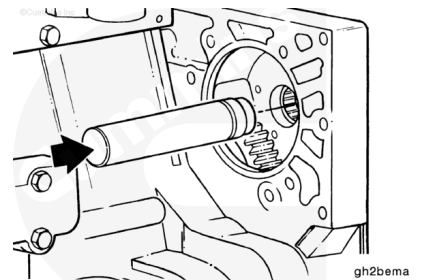
Remove the water pump. Twist the pump outward from the top, and angle the rear of the pump downward, as it is being removed, to allow the pump to pass the thermostat housing support.

Remove the water pump water transfer connection from the water pump.



Check the needle bearing for damage and freedom of needle rotation. Replace the bearing if it is damaged or if it does not turn freely.

If the bearing is replaced, use bearing installation tool, Part Number 3824117, along with a cup driver, to remove the needle bearing from the gear housing. Gently tap the bearing out from the rear side of the housing.

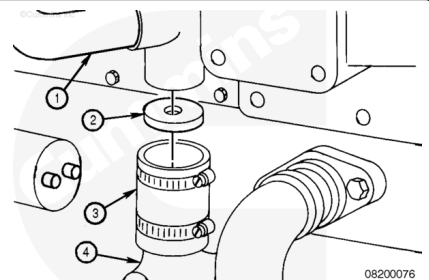


Without EGR

The coolant flow that provides cooling to the torque converter (if equipped) is achieved in different manners.

ISM and QSM Series engines use a torque converter cooler disc inside the coolant bypass hose to direct engine coolant to the inlet side of the torque converter cooler.

1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.

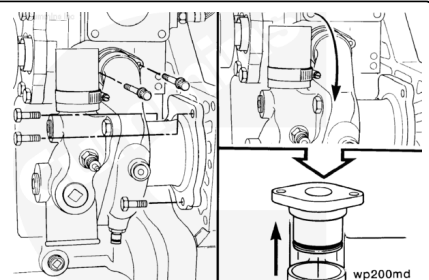


Remove the two water pump water transfer connection capscrews.

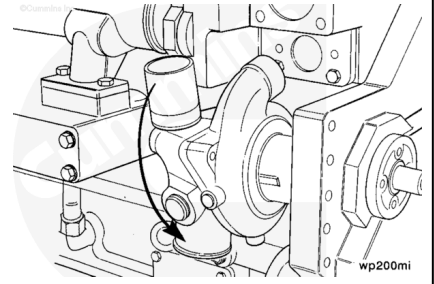
Remove the three water pump mounting capscrews.

Rotate the water pump outward so the water transfer connection can be removed from the water pump.

Remove the water transfer connection from the water pump.



Remove the water pump. Twist the pump outward from the top, and angle the rear of the pump downward, as it is being removed, to allow the pump to pass the thermostat housing support.



Check the needle bearing for damage and freedom of needle rotation. Replace the bearing if it is damaged or if it does **not** turn freely.

If the bearing is replaced, use bearing installation tool, Part Number 3824117, along with a cup driver to remove the needle bearing from the gear housing. Gently tap the bearing out from the rear side of the housing.

